SCOTT BERNETICH

Mechanical Design and Additive Manufacturing Engineer

Aberdeen, MD | 302-981-3992 | bernetichs001@gmail.com | linkedin.com/in/3dprint | Security Clearance: Secret

PROFESSIONAL SUMMARY

Innovative and results-driven Mechanical Engineer with 4+ years of hands-on experience spearheading Design for Additive Manufacturing (DfAM), CAD optimization, and end-to-end production across multi-material regimes (metals and polymers). Serving as one of just four Polymer AM Subject Matter Experts (SME's) for the U.S. Army Research Laboratory's Science of Extreme Materials Division, with exclusive ownership and technical command over all large-format polymer systems on base. Exceptional track record of accelerating product life cycles from conceptualization to fully deployment-ready, field-tested hardware for high-stakes defense, medical robotics, and commercial consumer applications

CORE TECHNICAL SKILLS

- CAD & Design: SolidWorks, Fusion 360, Blender, Creo, Design for Additive Manufacturing
- Large Format AM: BigRep Altra 670, Cincinnati MAAM, worked with: AES, Prototyp3, Caracol, CEAD
- Polymer AM: Stratasys, Formlabs, Markforged, Prusa, Bambu, and most entry level 3D Printer manufacturers
- Metal AM: EOS, Renishaw, 3D Systems
- Machining & Manufacturing: CNC mill (HAAS-trained), lathe, waterjet, 6-axis robotic arm programming
- Testing & Analysis: Design of Experiments, tensile/load-frame testing, Python, MATLAB, Minitab, Process Simulator
- Program & Delivery: budget/timeline ownership, technical reporting, mentorship, client & stakeholder management

WORK EXPERIENCE

Research Engineer — Additive Manufacturing & Polymer Processing | SURVICE Engineering, in support of the U.S. Army Research Laboratory — Aberdeen, MD *Jan 2024 – Present*

- Spearheaded expansion into Large-Format Additive Manufacturing (LFAM) by identifying critical capability gaps and orchestrating the end-to-end procurement, facility integration, and parameter qualification of industrial BigRep Altra 670 and Cincinnati MAAM systems; engineered custom metal AM/CNC post-machined ram extruders and conformal slot dies to reliably execute multi-kilogram, 100+ hour production runs for defense prototypes.
- Cultivated next-generation engineering talent by mentoring two military cadet interns in additive manufacturing; led them from project selection and SOW definition through design, testing, and final briefings, building hands-on AM expertise transferable to their service careers.
- Mitigated critical supply chain bottlenecks for deployed forces by partnering with unit commanders and product managers to redesign obsolete, high-cost communication hardware using Design for Additive Manufacturing (DFAM); developed hybrid metal-polymer reinforcement assemblies, mechanically qualified the parts, and deployed distributed digital technical data packages to forward operating bases from Kuwait to Alaska, rapidly restoring vital mission-critical equipment.
- Pioneered novel multi-material manufacturing capabilities by custom-retrofitting open-source hardware into an automated, Python-controlled 5-head hybrid FDM/Direct Ink Write (DIW) platform; formulated precise deposition workflows for rheologically complex fluids (shear-thickening, reactive, and energetic simulants) and integrated AI-assisted non-planar toolpath generation to expand advanced materials research.
- Engineered and scaled next-generation dual-material filament technology, designing custom extrusion dies and surface treatments to manufacture novel PC-cored, ABS-shrouded filament offering superior rigidity, simplified printability, and deformation-free annealing; successfully transitioned technical data packages to commercial manufacturing partners to supply warfighters at production scale.
- Accelerated aerodynamic and metallurgical research across joint services, collaborating with the USAF Innovation Lab to optimize high strength-to-weight structural components for a 6-foot fixed-wing UAS through flight testing, while concurrently establishing baseline sintering processes and as-printed applications for dense metal filaments in collaboration with multiple army groups.
- Directed industry partnerships, defining technical scopes of work (SOWs), contract solicitations, and milestone evaluations; authored high-visibility technical reports and proposal materials that directly contributed to SURVICE Engineering securing a competitive 5-year contract extension.

Lead Additive Manufacturing Engineer | Dive Design / 3D Pets — Boonton, NJ *Dec 2022 – Jan 2024*

- Directed end-to-end additive development for a dynamic portfolio of client products, matching functional requirements against the full AM spectrum to meet tight commercial launch windows.
- Consulted directly with client engineering and manufacturing teams to resolve complex design bottlenecks, applying DFAM principles to consolidate assemblies, reducing lead times, and transition parts from digital prototypes to shipped end-use inventory.

- Architected critical mechanical assemblies and additive production workflows for venture subsidiaries Willo and 3D Pets, engineering custom biomechanical animal prosthetics and mobility carts utilizing iPhone LiDAR 3D scanning and flexible TPU 3D printing—work recognized globally, including in Apple's "Made on iPhone" campaign; concurrently engineered artistic, ergonomic, and structural components for Willo's sustainable furniture portfolio.
- Collaborated with materials producers such as Braskem [Xtellar], PolySpectra, and HeyGears to test and qualify cutting-edge polymers then delivering full advertisements in the form of social media posts and videos.
- Served as technical authority and brand representative across major trade shows and additive industry events (e.g., Formnext, Additive Manufacturing Strategies), leading booth demonstrations, and translating complex multi-process capabilities to prospective clients.

Robotics Mechanical & Manufacturing Engineering Intern (2 rotations) | Johnson & Johnson — Robotics and Digital Solutions — Santa Clara, CA *May 2021 – Aug 2022*

- Designed test coupons for a Design of Experiments (DoE) study characterizing bond-line strength on a robotic-surgery stator bonding process
- Authored manufacturing process instructions and visual guides in SolidWorks Composer; trained in GD&T
- Simulated and validated a new manufacturing line via Process Simulator and cycle-time studies; led gap analysis across engineering, technician, and management stakeholders
- Designed and prototyped complex alignment fixtures to improve assembly-line throughput

Safety Manager | Penn State Learning Factory and Maker Space — State College, PA *Aug 2019 – Jun 2020*

- Trained students on mill, lathe, waterjet, and CNC equipment; oversaw safety compliance and led reorganization of the shop floor

FREELANCE & INDEPENDENT PROJECTS

- Founded and operated a freelance design practice, delivering production-ready 3D models for collectibles, jewelry, and custom hardware; managed the full client lifecycle (project scoping, scheduling, invoicing, and coordination with external manufacturers) to take designs from concept to finished product.
- Restored and upgraded a 2016 Honda motorcycle: repaired cracked fairings, replaced inoperable throttle assembly, mounted frame sliders, conducted routine maintenance, and designed a fully custom, 3D printed tail tidy with integrated lights and mounts.
- Designed and built multiple custom CoreXY and bed-slinger 3D printers from the ground up, integrating frame design, motion systems, electronics, and firmware; engineered application-specific extruders optimized for lightweight motion, high-flow throughput, and flexible materials, and maintained a fleet of 13 printers at its peak for supporting freelance production and hardware experimentation.
- Designed and built remote-controlled electromechanical systems from concept to operation, including an electric longboard exceeding 32 MPH, custom multirotor drones, and a competitive antweight (1 lb) combat robot; handled component selection, power and drive system integration, and fabrication for each platform.

EDUCATION

B.S., Mechanical Engineering — The Pennsylvania State University, 2022 (Minor: Classics & Ancient Mediterranean Studies)
Researched 3D-printed concrete applications at Penn State's CITEL Lab

LEADERSHIP & ACTIVITIES

- Officer, Vice President, President — Penn State 3D Printing Club (2018–2022): taught CAD/printer-repair lessons, organized design competitions and guest speakers
- Collegiate Sailor — Penn State Sailing Team (2018–2022)